

Qatar-2b

R-0979

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_170602 · created 2026-07-29T19:45:07+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457906.72494 +/- 0.00046 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.001 +/- 0.064
Transit depth (Rp/Rs)^2	0.00012 +/- 0.014 [%]
Orbital Inclination (inc)	89.77 +/- 1.72
Airmass coefficient 1 (a1)	3.48 +/- 1.55
Airmass coefficient 2 (a2)	-0.95 +/- 0.9
Scatter in the residuals of the lightcurve fit is	40.02 %
Best Comparison Star	#1 - [208, 199]
Optimal Aperture	2.53
Optimal Annulus	7.01
Transit Duration (day)	0.068 +/- 0.0029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.001 $\pm$ 0.064 vs 0.16327 (deviation 2.54 σ)
Duration fit vs published	0.068 d vs 0.0767 d (deviation 2.99 σ)
Mid-time O-C	-0.2 min
Depth SNR	0.0
Residual scatter	40.02 %

Light curve

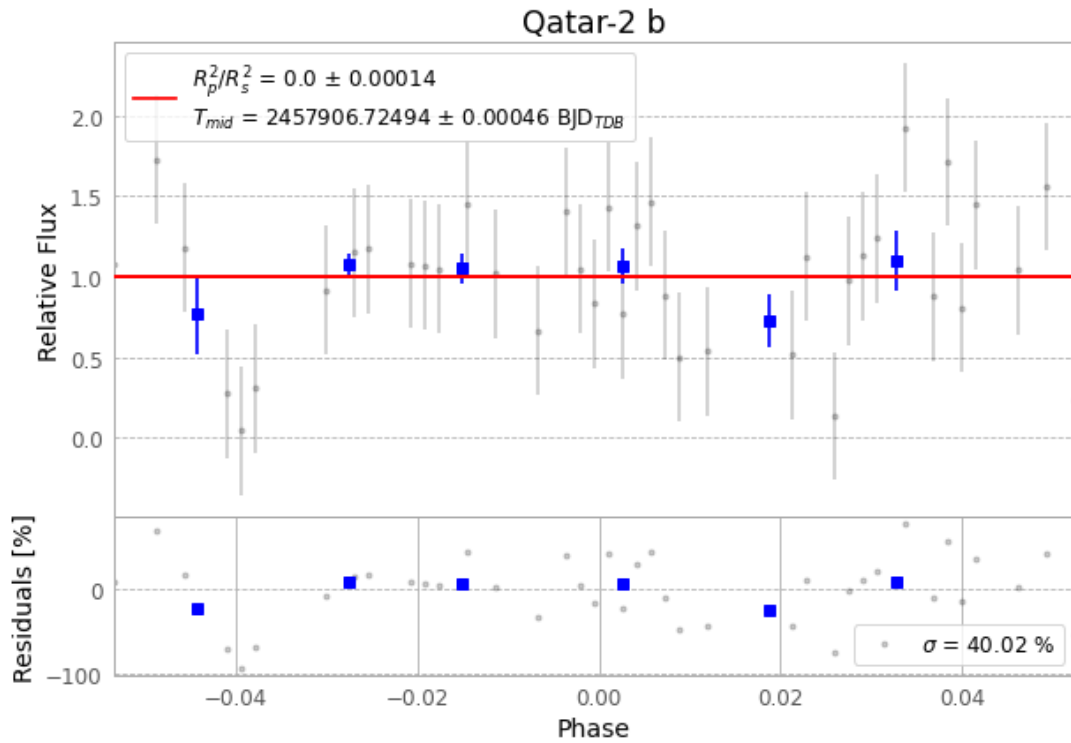


Figure 1 — FinalLightCurve_Qatar-2 b_2017-06-01.png (SHA-256 597777db12f2072d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [236, 356], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 68e0ec4409b8eec6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

70 FITS frames (46 MB), Qatar-2170602033312.FITS → Qatar-2170602070612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-170602.log (d453349f2df1...), FinalLightCurve_Qatar-2 b_2017-06-01.png (597777db12f2...), FinalLightCurve_Qatar-2 b_2017-06-01.csv (786c303481e1...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 19:45Z.